

Real-Time Weather Monitoring System using FPGA

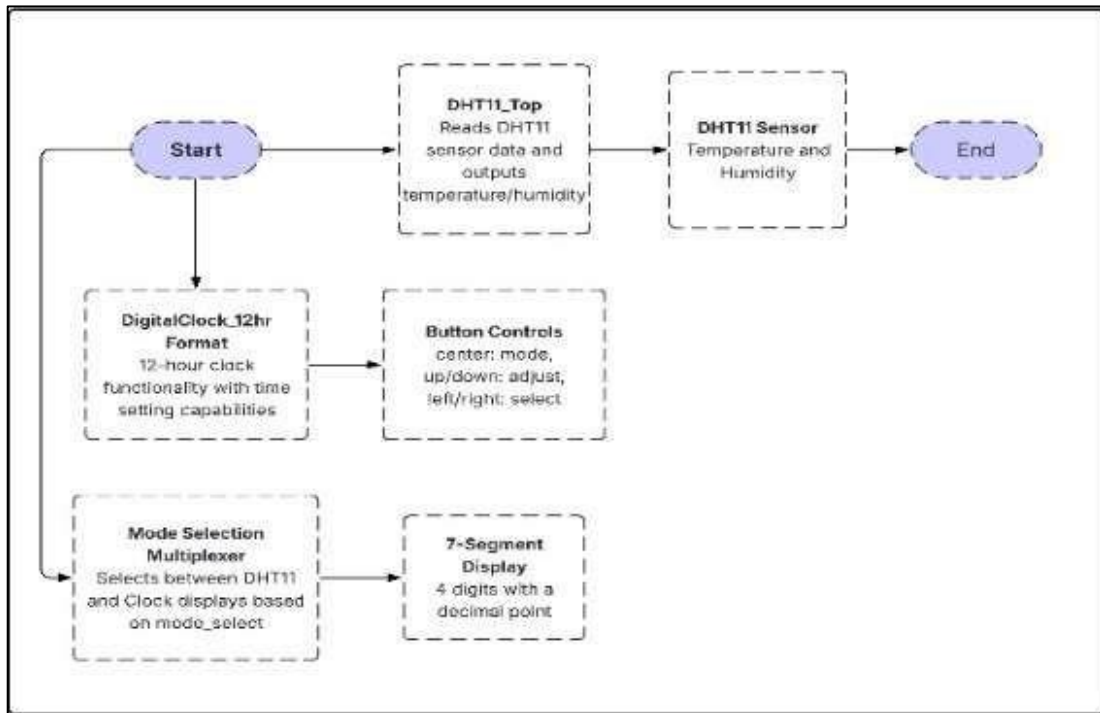


FIG.1 – Block Diagram

(4) RESULTS AND DISCUSSIONS



Fig.2 Default Clock setup

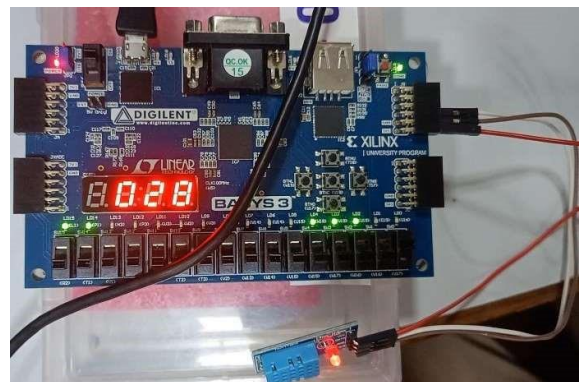


Fig.5 Real-time Temperature value using DHT11 sensor



Fig.3 Manual modification of Real-time Clock



Fig.4 Real-time Humidity value using DHT11 sensor

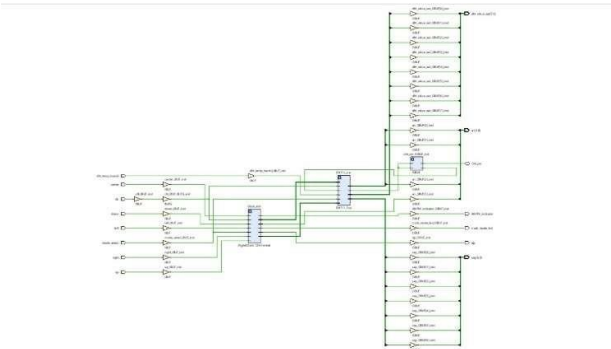


Fig.6 Schematic diagram

Name	^1	Slice LUTs (20800)	Slice Registers (41600)	Bonded IOB (106)	BUFGCTRL (32)
Integrated_Top		645	265	31	1
Clock_inst (DigitalClock_12hrFormat)		440	129	0	0
SSM (Seven_Segment_Module)		27	60	0	0
DHT11_inst (DHT11_Top)		205	136	0	0
DHT11_reader_inst (DHT11_Reader)		186	106	0	0
Seven_Segment_Display_inst (Sever		19	30	0	0

Fig.7 Utilization report

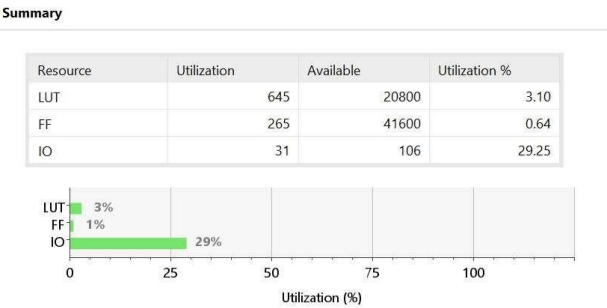


Fig.8 Utilization summary

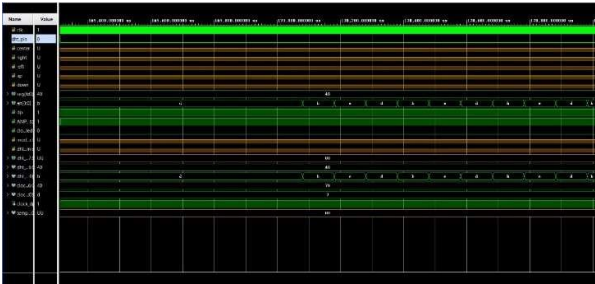


Fig.9 Simulation Output

FPGA Resource Utilization:

Resources	Total	Utilized	Percent
LUT	20,800	645	3.10
Flip-Flops	41,600	265	0.64
I/O Utilization	106	31	29.25